Dakshit Singh

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Education

SIT Pune
B.Tech in Computer Science & Engineering

Technical Skills

AI/Agentic Systems: LangGraph, CrewAI, OpenAI Agents SDK, PydanticAI, LangChain, RAG, LLMs, GenAI, Vector Databases, MCP, Function Calling
Backend & Languages: Python, JavaScript/TypeScript, C++, Dart, FastAPI, Node.js, Express, Flask, WebSockets, REST APIs, GraphQL, Microservices
Databases: PostgreSQL, MongoDB, Redis, Weaviate, MySQL, Firebase
DevOps & Cloud: Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Kafka, RabbitMQ, Prometheus, Grafana, Terraform, Ansible, AWS (EC2, S3, RDS, Lambda), Azure
Operating Systems: Windows, Linux (server admin, CLI workflows)
Other Tools: Git, Agile, Postman

Experience

BMC Software — **Software Engineer Intern**
Jan 2026 - Apr 2026 — ***Python, MCP, FastAPI, LLMs, Multi-Agent Systems***

- Built a **real-time, production-scale chatbot** for **Control-M** (BMC's flagship workload orchestration product), enabling natural-language control and observability directly from the web console.
- Architected an **MCP (Model Context Protocol) client-server** integration that exposed Control-M's automation APIs as tools, enabling **tool calling** and **multi-agent** workflows over enterprise orchestration data.
- Integrated the chatbot into the **BMC core application pool**, syncing with internal services and authentication to deliver a seamless, low-latency UX for Control-M operators.

Qlaws.ai – Legal Solution — **Backend Engineer**
Apr 2025 - Dec 2025 — ***LangGraph, Weaviate, Redis, PostgreSQL, Python***

- Refactored summarization pipeline with **LangGraph orchestration**, reducing processing time by **60%** and migrating **100K+ vectors** to Weaviate.
- Built a **multi-agent legal drafting system** automating LOI-to-SPA conversion using **6 specialized agents** for extraction, validation, and interaction.
- Designed a **dual-architecture chatbot** (Redis + PostgreSQL) handling **500+ concurrent** legal document queries with **sub-200ms** response times.

MahSoS Pvt Ltd. — **Backend Engineer**
Mar 2024 - Feb 2025 — ***Prisma, PostgreSQL, TypeScript, Node.js, Postman***

- Built and shipped backend features for the **Pram Edu** platform, owning end-to-end API design, implementation, and testing.
- Improved query performance via **indexing and query optimization** on PostgreSQL with Prisma ORM.
- Designed and validated REST APIs using Postman, ensuring reliable integration with frontend services.

DailyGossips (Freelance)
Oct 2023 - Dec 2023 — ***Flutter, Firebase, AWS S3, HLS***

- Developed a Marathi news app with **infinite scrolling**, real-time reactions, and engagement features.
- Integrated **HLS video streaming** with **AWS S3** storage and Firebase auth for scalable, low-latency delivery.

Projects

TTSaaS - VoiceAPI (ongoing project)
Designing a cost-optimized, high-throughput Text-to-Speech SaaS — ***Node.js, TypeScript, Python, Redis, Docker, AWS S3***

- Building a multi-tenant TTS SaaS that produces human-like voice audio at 5–10x lower cost by combining model distillation, batching, streaming inference, and spot-instance GPU autoscaling.
- Designed an event-driven pipeline using Express API → Redis/BullMQ queue → Python inference workers (Coqui/TTS or VITS) → S3 storage with presigned URLs for downloads/streaming.
- Implemented cost controls: dynamic batching, quantized/ONNX runtime models, GPU spot autoscaling, usage-based pricing tiers and caching of generated audio.

ChartMind AI
Autonomous Trading Agent with Chart-Reading & Live Execution — ***Python, TradingView, Broker APIs***

- Built an **automated trading agent** consuming **TradingView alerts** and executing trades on **leveraged CFDs** via broker APIs, with **Elliott Wave** state-machine pattern recognition for entries.
- Engineered risk management (position sizing, SL/TP, drawdown caps) and idempotent execution; **deployed live** with personal capital generating consistent returns.

YouTube Video Generation
Automated Video Generation and Uploading Pipeline — ***Linux, GPU, Backblaze, Multithreading***

- Architected a **fully automated YouTube pipeline** on Linux GPU servers handling script generation, voiceover synthesis, rendering, and scheduled uploads end-to-end.
- Leveraged **multithreading/multiprocessing** for parallel audio/video rendering, maximizing GPU utilization and throughput.
- Built scalable **batch processing** with fault tolerance and retries; used **Backblaze B2** for cost-efficient storage.